

Ultrathin Monolithic All-Nitride Ferroelectric Junctions for Cryogenic Memory and Compute-in-Memory Applications

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Abstract—We demonstrate cryogenic ferroelectric operation in an ultrathin monolithic all-nitride heterostructure junction based on 1 nm AlN / 4 nm AlScN / 1 nm AlN. To our knowledge, this is the first CMOS-voltage compatible cryogenic nitride ferroelectric junction and the thinnest cryogenic nitride ferroelectric device reported to date. The 6 nm junction shows stable polarization switching from 400 K to 14 K, low-voltage operation, enhanced breakdown, endurance stability, and stable 85 °C retention. Temperature- and field-dependent transport support a tunnel-dominant local trap-assisted readout mechanism, while DFT-informed trap energetics and local WKB modeling quantitatively support the measured current scale and field dependence. Finally, multilevel conductance programming enables measured-device-based vector-matrix multiplication for application in qubit error detection and decoding.

I. INTRODUCTION

Cryogenic quantum processors require local nonvolatile memory and compute near the qubit-control plane to reduce wiring density, latency, and room-temperature data transfer [1]. However, existing memory technologies face challenges in cryogenic operation, including high write voltage, selector/transistor overhead, limited scalability, or degraded reliability.

Wurtzite nitride ferroelectrics such as AlScN are promising because of their large polarization, thermal robustness, and sputter-based BEOL compatibility [2]. Yet prior cryogenic nitride ferroelectric demonstrations have mainly used thicker single-layer capacitor structures, which require higher voltage and do not provide tunnel-sensitive nonvolatile resistance readout [3,4]. For cryogenic compute-in-memory, an ultrathin heterostructure junction that combines polarization switching, reliable resistance readout, and interface-engineered breakdown is needed.

Here, we demonstrate a monolithic all-nitride ferroelectric junction based on a sputtered 1 nm AlN / 4 nm Al_{0.64}Sc_{0.36}N / 1 nm AlN stack. To the best of our knowledge, this is the first CMOS-voltage cryogenic nitride ferroelectric junction and the thinnest cryogenic nitride ferroelectric device reported to date. The device operates from 400 K to 14 K, shows tunnel-dominant cryogenic readout supported by DFT-informed WKB modeling, and enables multilevel conductance programming for measured-device-based vector-matrix multiplication and qubit error decoding.

II. DEVICE ARCHITECTURE AND CRYOGENIC FERROELECTRIC OPERATION

The device is a monolithic sputtered Al / AlN / AlScN / AlN / Al junction deposited without vacuum break and fabricated using a 300 °C BEOL-compatible process. The active stack is 1 nm AlN / 4 nm AlScN / 1 nm AlN, giving a total active thickness of only 6 nm. This ultrathin all-nitride heterostructure reduces the operating voltage by aggressive thickness scaling, while the symmetric AlN interfaces enhance breakdown robustness and form a tunnel-sensitive readout barrier, as shown in Fig. 2.

The 6 nm junction preserves ferroelectric operation from 400 K to 14 K. DC J–V, AC J–V, and C–V measurements confirm hysteretic switching and polarization-dependent electrostatic modulation in both 20 nm and 6 nm devices, while temperature-dependent coercive-field measurements show stable switching down to 14 K, as shown in Figs. 4–7. Notably, Fig. 7 shows that the coercive field becomes stabilized below 150 K, in contrast to the commonly expected monotonic increase of E_c upon cooling in AlScN FeCap operation [3]. This low-temperature stabilization suggests that switching in the ultrathin heterostructure is no longer governed solely by thermally activated domain nucleation. As thermal activation is progressively suppressed, the switching field may approach an interface- and field-controlled limit determined by the ultrathin AlScN layer, electrostatic boundary conditions, and symmetric AlN interfaces. This interpretation is consistent with the preserved hysteretic DC and AC switching from 400 K to 14 K in Figs. 4 and 5, the maintained capacitance hysteresis in Fig. 6, and the weak temperature dependence of E_c below approximately 150 K in Fig. 7. Such behavior prevents an excessive increase in programming voltage at cryogenic temperature and distinguishes the scaled heterostructure from thicker AlScN capacitors.

The scaled 6 nm device maintains an ON/OFF ratio larger than 2 across the full measured temperature range, with a 14 K coercive field of approximately 5.4 MV/cm from DC measurements. PUND measurements further verify switchable polarization: the 20 nm device shows saturated switching with an extracted $2P_r$ of 230 $\mu\text{C}/\text{cm}^2$, and the 6 nm device shows leakage-suppressed triangular PUND switching at 100 kHz, as shown in Figs. 8 and 9. The breakdown field reaches approximately 11.5 MV/cm under 10 kHz operation in Fig. 10, supporting reliable operation in the ultrathin heterostructure.

The reliability of the polarization and readout states is further confirmed in Fig. 11. The 6 nm junction shows long retention at 85 °C with clearly separated HRS and LRS states, while endurance measurements show that $2P_r$ remains nearly unchanged until hard breakdown. The ON/OFF ratio also remains stable during cycling, and the DC J–V evolution

indicates that repeated switching does not gradually collapse the resistance window before failure. These results verify that the ultrathin junction preserves both ferroelectric polarization and nonvolatile resistance readout under electrical stress.

The best device performance is summarized in **Fig. 3**. At 14 K, the 6 nm junction shows fast switching with a 50 ns FWHM pulse and an extracted switching energy density of 160 pJ/ μm^2 , corresponding to a projected approximately 0.2–0.26 pJ per cell at a 40 nm node [10]. The benchmark in **Fig. 3e** [3–9] further shows that this all-nitride platform provides large cryogenic polarization tuning at low voltage, distinguishing it from thicker nitride capacitors and oxide ferroelectric devices.

III. SIMULATION AND TRANSPORT MECHANISM

To identify the cryogenic readout mechanism, we measured temperature- and field-dependent DC transport in both resistance states of the 6 nm AlN/AlScN/AlN junction, using $E_{\text{avg}} = V/d$ as the average applied field. The read current does not follow a purely thermally activated mechanism. Instead, **Fig. 13** shows that both states approach a weakly temperature-dependent current floor at low temperature (<200 K), while the LRS gains additional thermal assistance at higher temperature. This indicates that cryogenic readout is dominated by tunneling with only weak thermal activation.

The mechanism is modeled as local trap-assisted tunneling through a polarization-modulated AlN/AlScN interface barrier. As illustrated in **Fig. 12**, polarization changes the local barrier profile, while the applied read field lowers and narrows the defect-saddle barrier, reducing the WKB escape distance to approximately 1–2 nm. Using an effective trap activation energy of approximately 0.7 eV, the same local WKB framework reproduces both the temperature-dependent current and the field-dependent sweep data in **Figs. 13** and **15**. DFT calculations further show that nitrogen-vacancy-related trap depths fall within this energy window, supporting the physical origin of the fitted trap level [11] in **Fig. 14a**.

Control comparisons rule out simpler transport pictures. Full-stack direct tunneling across the entire 6 nm barrier underestimates the measured current by approximately 15 orders of magnitude, while localized 1–2 nm barriers match the experimental current scale, as shown in **Fig. 14b**. Classical Poole–Frenkel emission is also insufficient, since the extracted field slope would imply an unphysical apparent permittivity far above the optical permittivity, although the measured field dependence remains PF-like in form. Therefore, the readout is best described as polarization-controlled local trap-assisted defect-saddle/WKB tunneling with weak thermal assistance, providing a mechanism consistent with the measured current scale, temperature dependence, and field dependence.

IV. MULTILEVEL CONDUCTANCE AND QUBIT-ERROR-DECODING VMM

Finally, we use the cryogenic conductance states of the 6 nm junction as programmable analog weights for vector–matrix multiplication. Continuous pulse programming produces reversible potentiation and depression, showing potential for

neuromorphic analog memory, while four-state retention confirms nonvolatile multibit conductance storage at a read voltage of -1.5 V, as shown in **Fig. 16**. Sequential up/down pulse trains further generate repeatable multilevel conductance trajectories, enabling measured-device-based weight mapping for analog VMM, as shown in **Fig. 17**.

To demonstrate relevance for cryogenic quantum control, the measured conductance states are mapped to a repetition-code syndrome decoder. Adjacent parity-check syndromes are encoded as VMM inputs, $x = [\text{bias}, s_1, s_2, \dots]$, and the programmed conductance matrix produces correction-output scores corresponding to no error or a single-qubit flip, as illustrated in **Fig. 18**. Signed weights are implemented using differential conductance pairs, $G^+ - G^-$, and measured run-to-run conductance variation is included with a 20% write-verify residual and 3% read noise. Importantly, the decoder uses only four measured conductance-difference levels, so the demonstration does not rely on high-precision multibit storage. The matrix is programmed once, frozen, and repeatedly used for syndrome decoding, matching the intended nonvolatile cryogenic memory operation.

Using measured conductance data, the decoder maintains high accuracy from 3 to 11 data qubits. At 3% physical bit-flip probability, the VMM decoder achieves about 99% any-error detection and about 97% zero/single-error decoding for 3 data qubits, while the correction success remains above 84% up to 11 data qubits, as shown in **Fig. 19**. These results show that the ultrathin all-nitride junction can provide not only cryogenic nonvolatile memory, but also multilevel analog compute for local qubit-error detection and decoding near the control plane.

V. CONCLUSION

We demonstrate an ultrathin, monolithic nitride ferroelectric junction for CMOS-voltage cryogenic memory and compute-in-memory. The 1 nm AlN / 4 nm AlScN / 1 nm AlN heterostructure preserves ferroelectric operation from 400 K to 14 K, with large polarization switching, enhanced breakdown, stable retention/endurance, and tunnel-dominant resistance readout. Using measured conductance states, we further demonstrate low-precision analog VMM for qubit-error detection and decoding. This work establishes all-nitride ferroelectric junctions as a promising BEOL-compatible platform for scalable cryogenic memory and quantum computing accelerators.

Acknowledgement: D.J. and Z.H. acknowledge support from ONR (NO. N00014–24–1–2131), AFOSR (NO. FA9550–23–1–0391), and NSF FuSe (2328743). A.A. and M.C. acknowledge support from REU summer research program. A.A. thanks Dr. Igor Mazin for computational resources.

References: [1] H. Bohuslavskyi et al., SNW, 2017. [2] S. Fichtner et al., J. Appl. Phys. 125, 114103, 2019. [3] S. Song et al., arXiv:2503.19491, 2025. [4] R. Wang et al., EDL, 2025. [5] T.-E. Lee et al., IEDM, 2023. [6] S. G. Kirtania et al., IEDM, 2024. [7] H. Bohuslavskyi et al., AEM, 2024. [8] S.-H. Kuk et al., IEEE TED 70, 4122, 2023. [9] J. Hwang et al., Nano Conver. 11, 58, 2024. [10] Z. Hu et al., Nano Lett. 25, 13748–13756, 2025. [11] Chen et al., Materials, 2024.

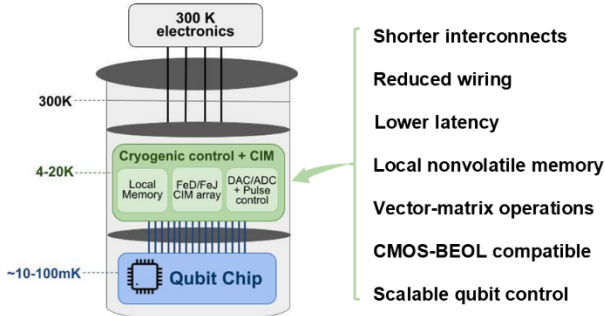


Fig. 1. System-level motivation: local cryogenic nonvolatile memory and vector-matrix operations near the qubit-control plane reduce interconnect density, latency, and room-temperature data transfer.

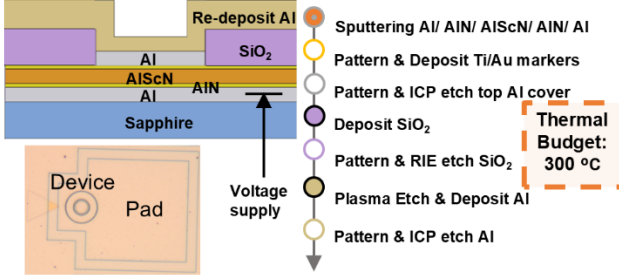


Fig. 2. Fabrication flow of monolithic all-nitride FTJ devices. The Al/AlN/AIScN/AlN stack is sputtered sequentially without vacuum break.

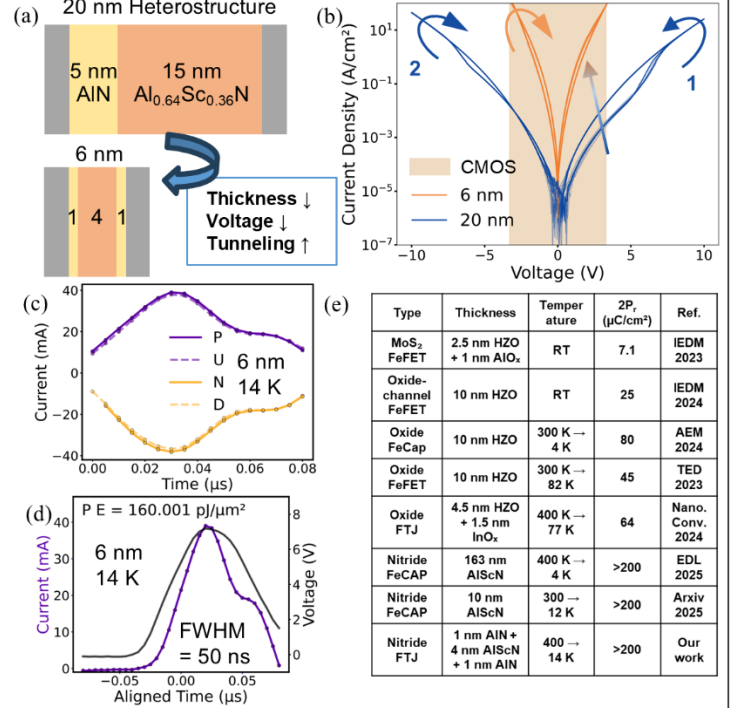


Fig. 3. (a) Device scaling. (b) DC J-V showing CMOS-compatible operation in the scaled junction. (c) Fast cryogenic PUND response. (d) Zoomed P pulse used to extract switching energy. (e) Benchmark comparison highlighting the ultrathin nitride FTJ.

Stable Ferroelectric Operations from 400 K to 14 K

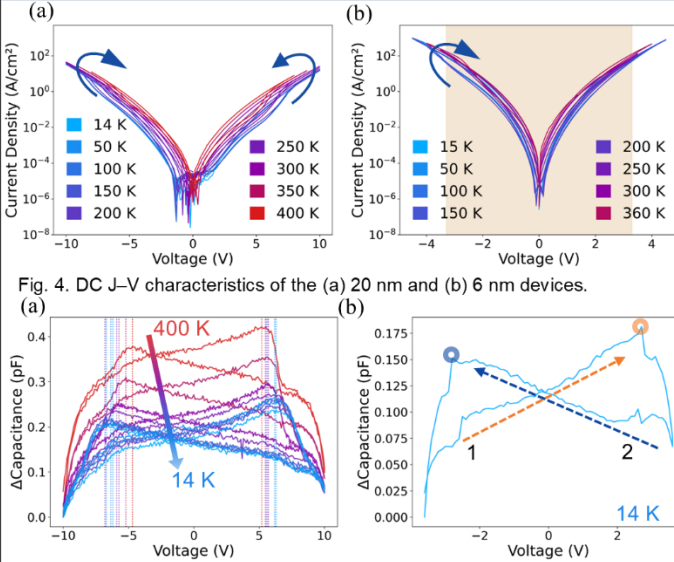


Fig. 4. DC J-V characteristics of the (a) 20 nm and (b) 6 nm devices.

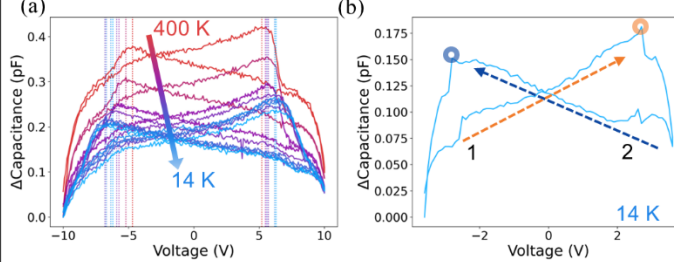


Fig. 6. C-V characteristics measured with a 100 kHz small-signal excitation on the (a) 20 nm and (b) 6 nm devices. Capacitance is calculated relative to the initial minimum value to minimize variations from the via structure.

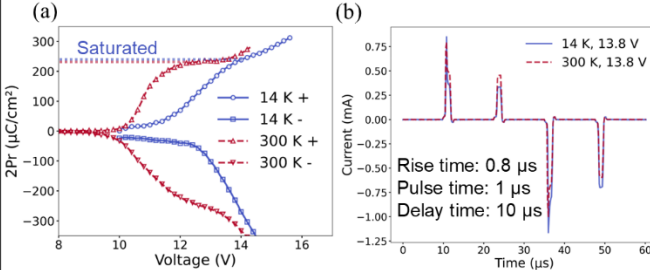


Fig. 8. PUND switching of the 20 nm device. (a) PUND polarization loops measured with gradually increasing voltage. (b) Raw PUND pulse data measured at 13.8 V.

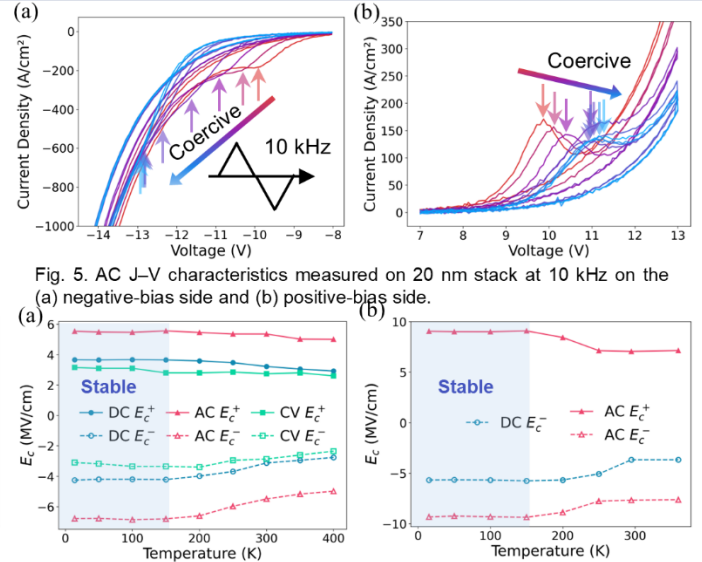


Fig. 5. AC J-V characteristics measured on 20 nm stack at 10 kHz on the (a) negative-bias side and (b) positive-bias side.

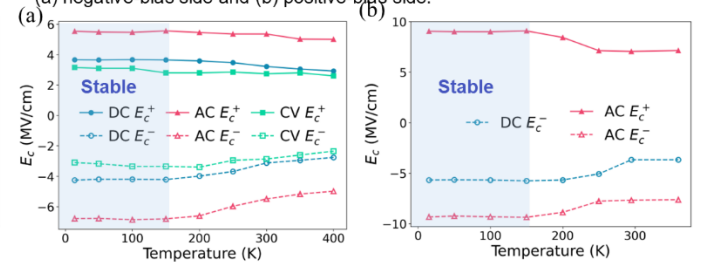


Fig. 7. Coercive field as a function of temperature for the (a) 20 nm and (b) 6 nm devices, showing stable operation from 14 K to 150 K.

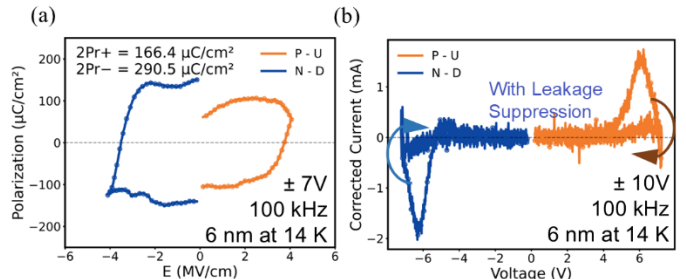


Fig. 9. Triangular PUND switching of the 6 nm device. Triangular PUND measurement at 100 kHz with leakage-current suppression. (a) P-E loop measured at 7 V. (b) Corrected current response measured at 10 V, where the current peak represents polarization switching.

Reliability and Breakdown Analysis

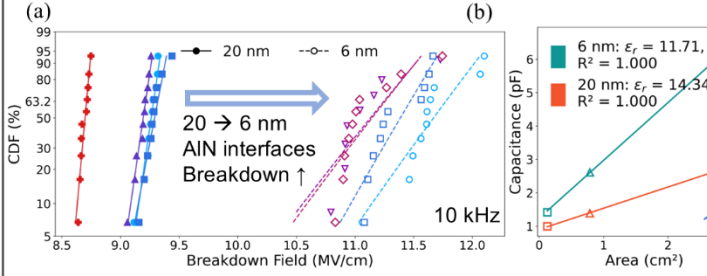


Fig. 10. Breakdown and capacitance analysis. (a) Weibull breakdown distributions vs. temperature, showing AlN-interface-enhanced breakdown in 6 nm devices. (b) Area-dependent capacitance for 14 K permittivity extraction.

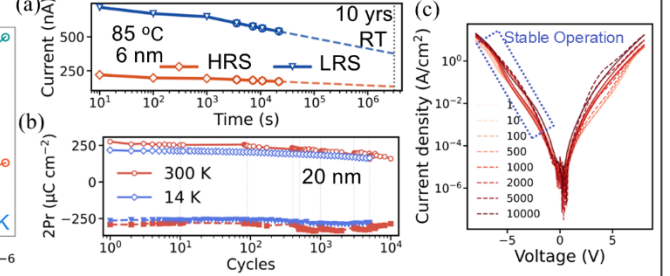


Fig. 11. Reliability. (a) 85 °C retention of the 6 nm device, showing distinct HRS/LRS states. (b) Endurance of the 20 nm device at 300 K and 14 K. (c) DC J-V evolution during cycling, showing stable transport after repeated switching.

Mechanism: Local Trap-Assisted Tunneling

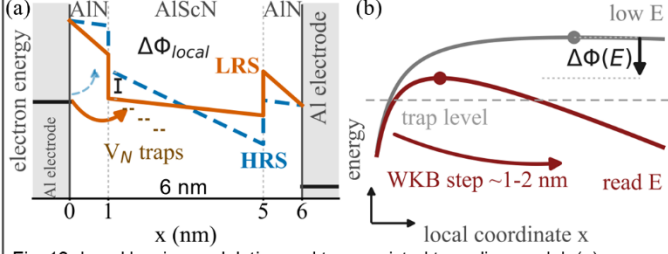


Fig. 12. Local barrier modulation and trap-assisted tunneling model. (a) Polarization-dependent barrier modulation at the AlN/AlScN read interface. (b) Field-thinned defect saddle enabling short-range WKB tunneling.

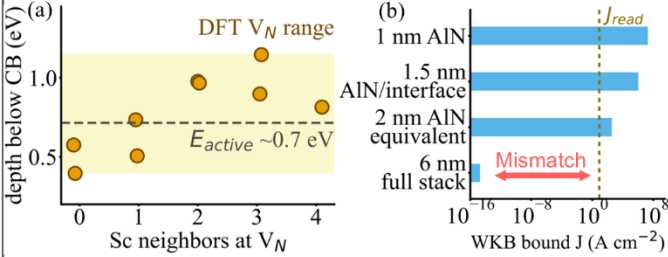


Fig. 14. Trap energy and local WKB current scale. (a) DFT V_N trap-depth window for different Sc-neighbor configurations, with $E_{active} \sim 0.7$ eV lying within the DFT-supported range. (b) WKB current bounds showing local AlN/interface barriers approach measured J_{read} , while full-stack direct tunneling is negligible.

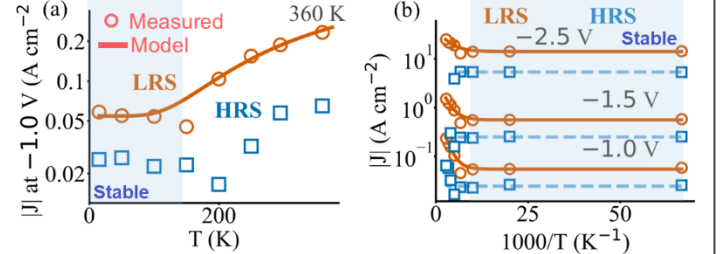


Fig. 13. Temperature-dependent read current. (a) Read current vs. temperature, showing a low-temperature floor with weak thermal assist. (b) Arrhenius plots at different read fields, confirming the floor-plus-assist behavior.

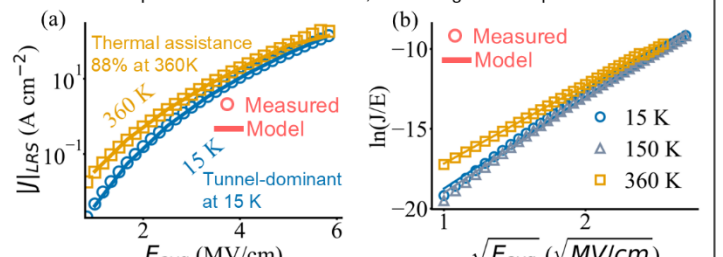


Fig. 15. Field-dependent validation. (a) LRS field validation at 15 K and 360 K. (b) PF-linearized field scaling, showing PF-like field scaling behavior.

Pre-trained Vector Matrix Multiplication for Qubit Error Correction

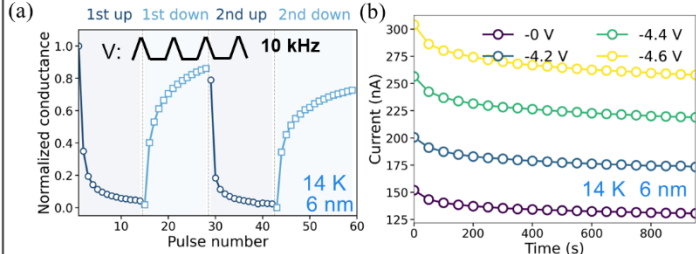


Fig. 16. Multilevel conductance modulation. (a) Continuous-pulse conductance update read at -1.5 V, emulating synaptic potentiation/depression. (b) Four-state multibit retention under different write biases, read at -1.5 V.

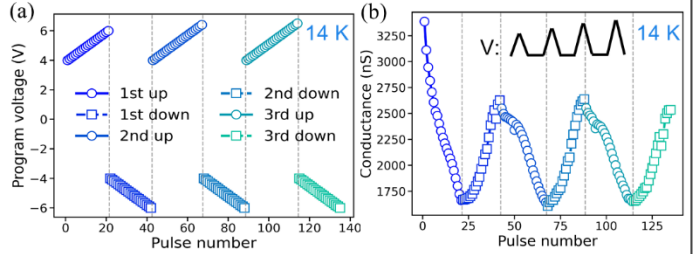


Fig. 17. Pre-trained conductance programming. (a) Sequential up/down program-voltage pulses. (b) Multilevel conductance response read at -1.5 V for analog VMM weight mapping.

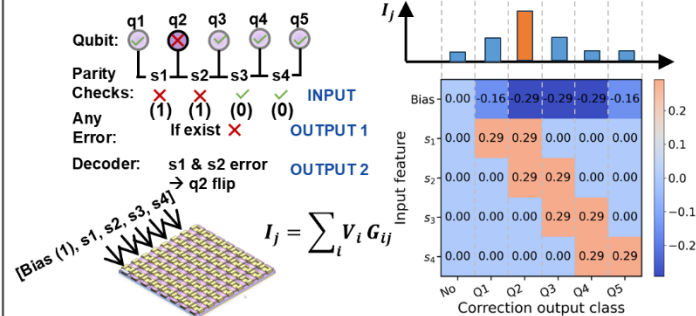


Fig. 18. Ferroelectric-array implementation of qubit error-correction decoding. Parity-check syndromes are encoded as VMM inputs and mapped through the programmed conductance matrix to generate correction outputs, enabling hardware-efficient qubit-error decoding.

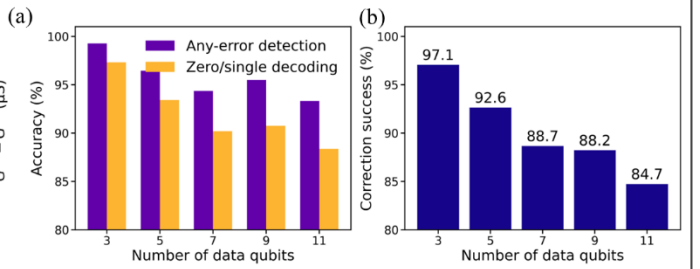


Fig. 19. Qubit error-correction performance using measured conductance. (a) Any-error detection and zero/single-error decoding accuracy versus qubit number. (b) Overall correction success decreases with code size but remains above 84% for up to 11 data qubits.